Guidelines for Running MM/PBSA Analysis Using gmx_MMPBSA

This document provides a step-by-step guide to perform MM/PBSA binding free energy calculations using gmx_MMPBSA after completing a molecular dynamics (MD) simulation in GROMACS.

Prerequisites

File	Purpose
prod.tpr	Production run topology
prod_fit.xtc	Fitted MD trajectory
topol.top	GROMACS topology
index.ndx	Index file containing Protein and Ligand groups
mmpbsa.in	MM/PBSA input parameter file

Step 1: Navigate to the Simulation Directory

cd ~/Gromacs_sim

Step 2: Verify Required Files

ls

Step 3: Activate the gmx_MMPBSA Environment

conda activate gmxMMPBSA

Step 4: Verify Installation

gmx_MMPBSA --version

Step 5: Run MM/PBSA Calculation

gmx_MMPBSA \
-O \
-i mmpbsa.in \
-cs prod.tpr \

```
-ct prod_fit.xtc \
-cp topol.top \
-ci index.ndx \
-cg 1 13 \
-o FINAL_RESULTS_MMPBSA.dat \
-eo FINAL_RESULTS_MMPBSA.csv
```

Explanation of Parameters

Option	Description
-O	Overwrite existing output files
-i	MM/PBSA input file
-cs	Production structure (.tpr)
-ct	Fitted trajectory (.xtc)
-cp	Topology file
-ci	Index file
-cg 1 13	Protein and ligand group numbers
-o	Output DAT file
-eo	Output CSV file

Step 6: Verify Output Files

```
ls FINAL_RESULTS*
```

Step 7: View the Results

```
cat FINAL_RESULTS_MMPBSA.dat
```

Expected Output Files

- FINAL_RESULTS_MMPBSA.dat
- FINAL_RESULTS_MMPBSA.csv
- gmx_MMPBSA.log
- COM.prmtop
- REC.prmtop
- LIG.prmtop

Workflow Summary

Go to simulation folder
↓
Check files (ls)
↓
Activate gmxMMPBSA environment
↓
Check software version
↓
Run gmx_MMPBSA
↓
Generate output files
↓
View FINAL_RESULTS_MMPBSA.dat
↓
Interpret Binding Free Energy

Commands Summary

```
cd ~/Gromacs_sim

ls

conda activate gmxMMPBSA

gmx_MMPBSA --version

gmx_MMPBSA \
-O \
-i mmpbsa.in \
-cs prod.tpr \
-ct prod_fit.xtc \
-cp topol.top \
-ci index.ndx \
-cg 1 13 \
```

```
-o FINAL_RESULTS_MMPBSA.dat \
-eo FINAL_RESULTS_MMPBSA.csv

ls FINAL_RESULTS*

cat FINAL_RESULTS_MMPBSA.dat
```